

Problems Used

Heat-1D

$$\begin{aligned}u(x, t) &= \sin(\pi x) \sin(\pi t) \\u_0(x) &= 0 \\u_t(x, t) &= \pi \sin(\pi x) \cos(\pi t) \\u_{xx}(x, t) &= -\pi^2 \sin(\pi x) \sin(\pi t) \\f(x, t) &= \pi \sin(\pi x) \cos(\pi t) + \pi^2 \sin(\pi x) \sin(\pi t) \\&= \pi \sin(\pi x) [\cos(\pi t) + \pi \sin(\pi t)]\end{aligned}$$

Heat-1D-Oscillating

$$u(x, t) = \sin(10\pi x) \sin(\pi t), \quad u_0(x) = 0$$

Same f as above, but with constant,

$$\text{so } f(x, t) = \pi \sin(10\pi x) [\cos(\pi t) + 100\pi \sin(\pi t)]$$

Heat-1D-Oscillating-Cosine

$$\begin{aligned}u(x, t) &= \sin(10\pi x) \cos(10\pi t) \\u_0(x) &= \sin(10\pi x) \\u_t(x, t) &= -10\pi \sin(10\pi x) \sin(10\pi t) \\u_{xx}(x, t) &= -100\pi^2 \sin(10\pi x) \cos(10\pi t) \\f(x, t) &= \sin(10\pi x) [100\pi^2 \cos(10\pi t) - 10\pi \sin(\pi t)]\end{aligned}$$

Linear-Heat-2d (default)

$$u(x, y, t) = e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

$$u_0(x, y) = \sin(\pi x) \sin(\pi y)$$

$$u_t(x, y, t) = (1 - 2t)e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

$$\begin{aligned} \Delta u &= e^{t-t^2} [-\pi^2 \sin(\pi x) \sin(\pi y) - \pi^2 \sin(\pi x) \sin(\pi y)] \\ &= -2\pi^2 e^{t-t^2} \sin(\pi x) \sin(\pi y) \end{aligned}$$

$$f(x, y, t) = (1 - 2t)e^{t-t^2} \sin(\pi x) \sin(\pi y) + 2\pi^2 e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

Linear-Heat-2D (cosrinsin)

$$u(x, y, t) = \sin(5\pi x) \sin(5\pi y) \sin(5\pi t)$$

$$u_0(x, y) = 0$$

$$u_t(x, y, t) = 5\pi \sin(5\pi x) \sin(5\pi y) \cos(5\pi t)$$

$$\Delta u = -50\pi^2 \sin(5\pi x) \sin(5\pi y) \sin(5\pi t)$$

$$f(x, y, t) = \sin(5\pi x) \sin(5\pi y) [5\pi \cos(5\pi t) + 50\pi^2 \sin(5\pi t)]$$

Nonlinear-Heat-2D

$$u_t - \Delta u + u^2 = f$$

$$u(x, y, t) = e^{t-t^2} \sin(\pi x) \sin(\pi y), \quad u_0 = \sin(\pi x) \sin(\pi y)$$

$$u_t(x, y, t) = (1 - 2t)e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

$$\Delta u = -2\pi^2 e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

$$u^2 = e^{2(t-t^2)} \sin^2(\pi x) \sin^2(\pi y)$$

$$f(x, y, t) = (1 - 2t)e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

$$+ 2\pi^2 e^{t-t^2} \sin(\pi x) \sin(\pi y)$$

$$+ e^{2(t-t^2)} \sin^2(\pi x) \sin^2(\pi y)$$